

NbTaV-(Ti,W) refractory high-entropy alloys: Experiments and modeling

H.W. Yao^a, J.W. Qiao^{a,*}, M.C. Gao^{b,c,**}, J.A. Hawk^b, S.G. Ma^d, H.F. Zhou^a, Y. Zhang^e

^a Laboratory of Applied Physics and Mechanics of Advanced Materials, College of Materials Science and Engineering, Taiyuan University of Technology, Taiyuan 030024, China

^b National Energy Technology Laboratory, 1450 Queen Ave SW, Albany, OR 97321, USA

^c AECOM, P.O. Box 1959, Albany, OR 97321, USA

^d Institute of Applied Mechanics and Biomedical Engineering, Taiyuan University of Technology, Taiyuan 030024, China

^e State Key Laboratory for Advanced Metals and Materials, University of Science and Technology Beijing, Beijing 100083, China

ARTICLE INFO

Article history:

Received 16 February 2016

Received in revised form

24 July 2016

Accepted 25 July 2016

Available online 26 July 2016

Keywords:

High-entropy alloys

Refractory

Single-phase

Design

Ductile

ABSTRACT

This study reports the design and development of ductile and strong refractory single-phase high-entropy alloys (HEAs) for high temperature applications, based on NbTaV with addition of Ti and W. Assisted by CALPHAD modeling, a single body-centered cubic solid solution phase was experimentally confirmed in the as-cast ingots using X-ray diffraction and scanning electron microscopy. The observed elemental segregation in each alloy qualitatively agrees with CALPHAD prediction. The Vickers micro-hardnesses (and yield strengths) of the alloys are about 3 (and 3.5–4.4) times that those estimated from the rule of mixture. While NbTaTiVW shows an impressive yield strength of 1420 MPa with fracture strain of 20%, NbTaTiV exhibits exceptional compressive ductility at room temperature.

© 2016 Elsevier B.V. All rights reserved.

1. Introduction

Structural engineering materials with high strength, acceptable ductility, and good thermal stability are highly desirable for critical applications [1,2]. Conventional alloys have one principal element as the matrix, with minor modifications achieved by adding relatively small amounts of other alloying elements. This strategy typically optimizes a set of physical or mechanical properties while retaining the base characteristic properties of the main alloying matrix element. Over the past decade, a new alloy design and development philosophy—high-entropy alloys (HEAs), or multi-principle-element alloys (MPEAs)—has attracted significant attention [3,4]. Such alloys generally consist of four, or more elements with equimolar or near-equimolar ratios that form face-centered cubic (FCC) [4], body-centered cubic (BCC) [5], or hexagonal close-packed (HCP) phases [6,7]. A shared rationale is to maximize the configurational entropy by increasing the number of the principal components, thereby, in theory, continuing to improve the thermal stability of disordered solid-solution phases against intermetallics, especially at elevated

temperatures. A number of HEAs have showed a variety of favorable, and sometimes unusual properties such as high strength, high ductility, high fracture toughness at low temperatures, and thermal stability. (For recent comprehensive reviews on HEAs, the reader is referred to Refs. [8,9]).

Traditional refractory metals and alloys hold great potential in high temperature applications beyond the capability of the state-of-the-art Ni-based superalloys, primarily due to their extremely high melting points, but they also suffer from low ductility at low temperatures and their oxidation resistance is poor [10]. Recent development on refractory HEAs suggests that they may exhibit outstanding strength [11] and exceptional oxidation resistance [12]. The main objective of the present work is to explore the design potential of single-phase refractory HEAs, namely, NbTaTiV, NbTaVW, and NbTaTiVW. These HEAs possess the BCC structure, resulting in alloys that are ductile and strong. The base alloy NbTaV was chosen since elements Nb, Ta, and V, separately, show excellent ductility at room temperature [13].

2. Computational methodologies and experimental procedures

The CALPHAD calculations were performed using the TCNI7 thermodynamic database supplied by ThermoCalc™ [14]. The

* Corresponding author.

** Corresponding author at: National Energy Technology Laboratory, 1450 Queen Ave SW, Albany, OR 97321, USA.

E-mail addresses: qiaojunwei@gmail.com (J.W. Qiao), michael.gao@netl.doe.gov (M.C. Gao).

database covers the complete binaries pertaining to nine refractory metals although it was initially developed for Ni-based superalloys. Previous CALPHAD modeling using this database demonstrated good agreement with experimental observations for various refractory BCC HEAs [15–17], but future development including descriptions for all possible ternary alloys in their complete composition ranges will definitely enhance the reliability of the database in the applications for HEAs. For guidance on CALPHAD modeling of HEAs, the reader is referred to recent publications [18–21].

Alloy ingots with nominal compositions of NbTaTiV, NbTaVW, and NbTaTiVW (compositions in at.%) were synthesized by vacuum arc melting of the equimolar pure metals (weight percentage purity $\geq 99.9\%$) in a Ti-gettered, high-purity argon atmosphere. Arc melting of the alloys was conducted on a water-cooled copper hearth crucible. To achieve as homogeneous distribution of elements as possible in the alloys using this melting technique, the ingots were remelted at least seven times, flipping them over between each melt. The resulting crystal structures developed in the solidified ingots were identified using X-ray diffraction. Cu K α radiation was used through the 2θ range between 20° and 80° . The morphologies and microstructures of the as-cast alloys were

characterized by scanning electron microscopy (SEM) with energy dispersive spectroscopy (EDS) capability. Vickers microhardness was used to determine the alloy hardness. An indent was placed on the polished cross-section surface, using a 136° Vickers diamond pyramid under 1 kg load, applied for 15 s. Cylindrical specimens for compression testing were electric-discharged machined from the cast ingots. The samples were 3 mm in diameter and 6 mm in height giving an aspect ratio of 1:2. The two end surfaces of the cylindrical compression specimen were carefully polished to obtain mirror-like surfaces as well as to insure planar and parallel sided surfaces. A MTS 809 universal testing machine was used to perform the compression tests at room temperature using a nominal strain rate of $5 \times 10^{-4} \text{ s}^{-1}$.

3. Results

3.1. CALPHAD modeling

The predicted equilibrium phase mole fraction versus temperature plots for these alloys are shown in Fig. 1. The calculated liquidus temperature (T_{liq}), solidus temperature (T_{sol}), and

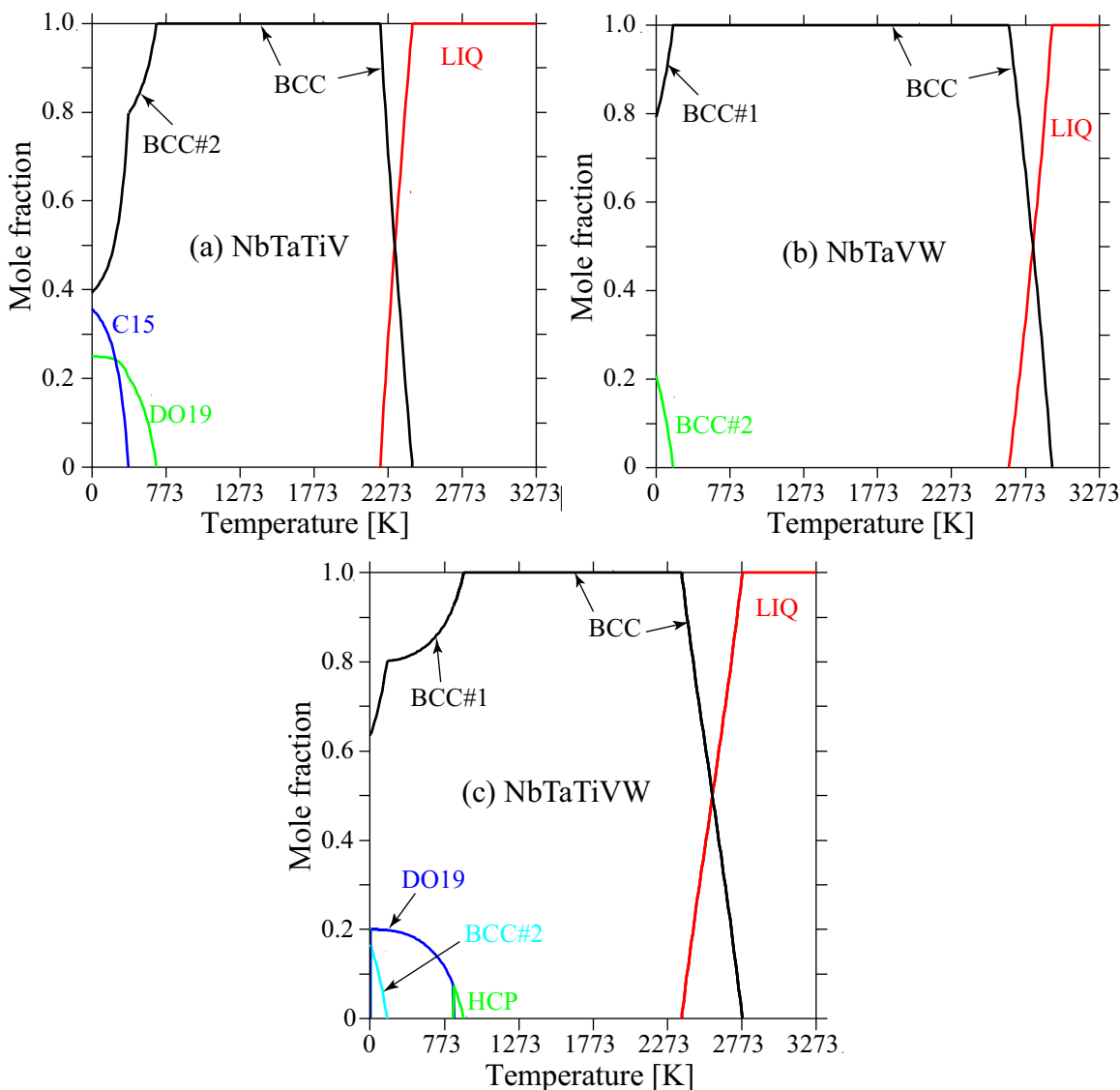
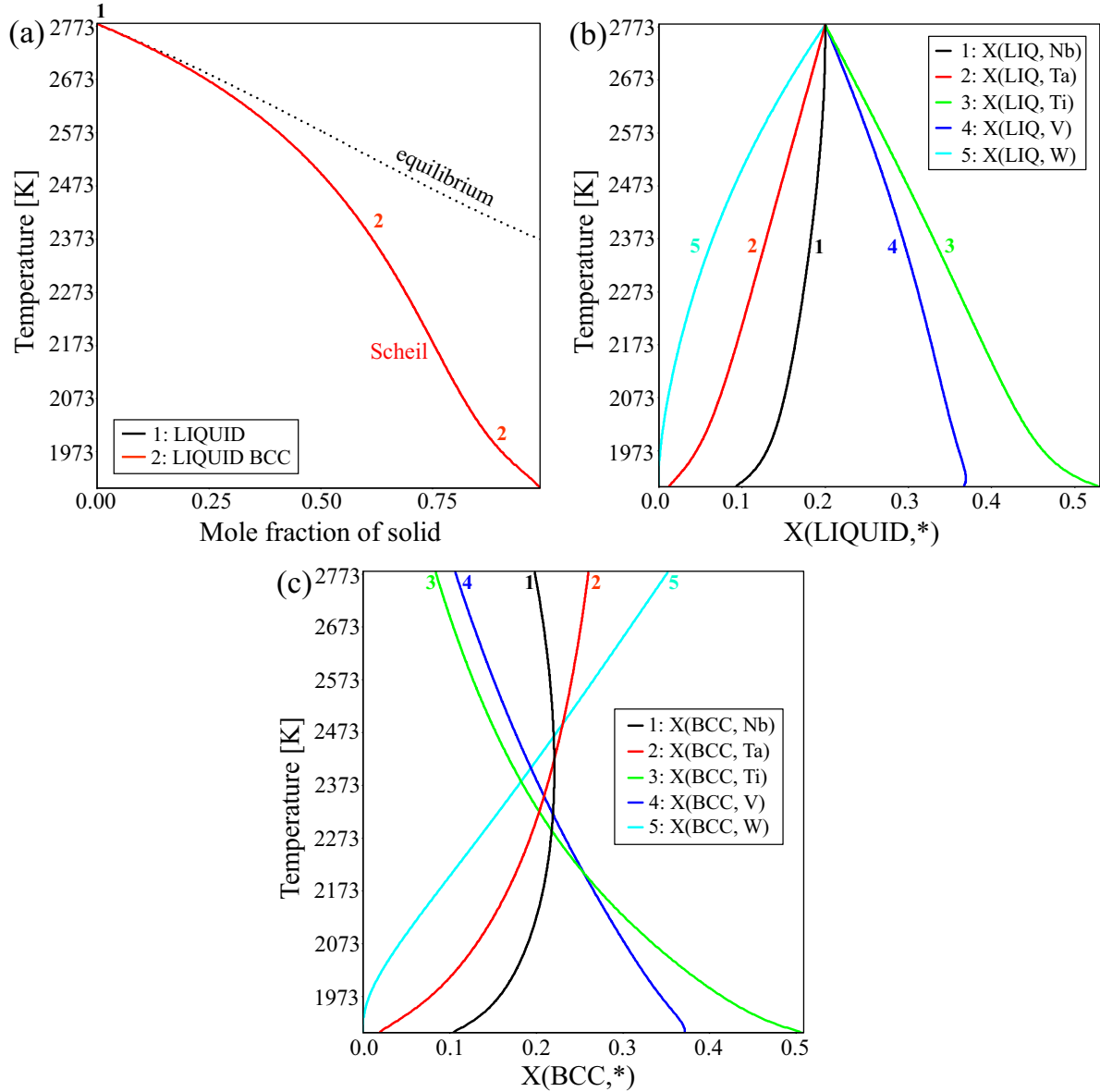


Fig. 1. Calculated equilibrium mole fraction versus temperature for HEAs: (a) NbTaTiV, (b) NbTaVW, and (c) NbTaTiVW.

Table 1

Predicted liquidus, solidus, and decomposition temperatures and mixing thermodynamic properties using TCNI7 database.

Alloys	T_{liq} (K)	T_{sol} (K)	T_{dec} (K)	$\Delta H_{\text{mix}}^{\text{liq}}$ (kJ/mol)	$\Delta H_{\text{mix}}^{\text{bcc}}$ (kJ/mol)	$\Delta S_{\text{mix}}^{\text{liq}}$ (J K ⁻¹ mol ⁻¹)	$\Delta S_{\text{mix}}^{\text{bcc}}$ (J K ⁻¹ mol ⁻¹)
NbTaTiV	2411	2177	709	−0.1	+3.2	11.5	11.6
NbTaVW	2953	2662	388	−7.2	−5.4	9.85	9.6
NbTaTiVW	2777	2369	835	−3.4	−1.8	12.3	11.7

Note: the reference states used are the liquid and BCC phases to calculate $\Delta H_{\text{mix}}^{\text{liq}}/\Delta S_{\text{mix}}^{\text{liq}}$ and $\Delta H_{\text{mix}}^{\text{bcc}}/\Delta S_{\text{mix}}^{\text{bcc}}$, respectively.**Fig. 2.** Non-equilibrium solidification of HEA NbTaTiVW using Scheil model: (a) mole fraction of solid, (b) liquid composition, and (c) BCC composition.

decomposition temperature (T_{dec}) of the BCC phase are listed in Table 1. HEA NbTaVW shows the greatest thermal stability since it possesses the highest T_{liq} and T_{sol} , as well as the lowest T_{dec} , followed by NbTaTiVW and then NbTaTiV.

The non-equilibrium solidification simulation was accomplished using the Scheil-Gulliver model [22,23], which assumes equilibrium mixing in the liquid and no diffusion in the solid. The simulations predict formation of a single BCC solid solution during the entire non-equilibrium solidification for the three alloys. An

example is shown in Fig. 2 for HEA NbTaTiVW. The real solidification path will lie somewhere between equilibrium and Scheil model (Fig. 2(a)). The simulation also predicts likely segregation tendency due to constitutional cooling. As the temperature decreases during solidification, the liquid is gradually depleted in W followed by Ta. Alternatively, it is steadily enriched in Ti followed by V (Fig. 2(b)). In contrast, the BCC phase is predicted to be rich in W and Ta in the dendrites that form at earlier stages of solidification. At latter stages of solidification, the BCC phase is enriched

in Ti and V in interdendritic regions (Fig. 2(c)).

3.2. Crystal structure

X-ray diffraction (XRD) patterns of the as-cast samples are shown in Fig. 3. The XRD peaks of the alloys are indexed as single phase BCC. The lattice parameters were determined to be 3.2193 Å, 3.2186 Å, and 3.2244 Å, respectively, for NbTaTiV, NbTaVW, and NbTaTiVW (see Table 2). These values agree very well with those estimated from the rule of mixture (ROM) [24]:

$$a_{\text{mix}} = \sum_{i=1}^n c_i a_i, \quad (1)$$

where c_i and a_i are the atomic fraction and the lattice parameter of the i^{th} element, respectively, and n is the number of alloy components. The lattice parameters of pure elements, taken from Ref. [13], are given in Table 2.

3.3. Microstructure and chemical compositions

Backscattered-electron SEM images of the investigated alloys are presented in Fig. 4. The dendritic microstructure is clearly seen in these alloys, which is typical of as-cast ingots. The dendrite arms appear to be “brighter” (in terms of SEM contrast) than the interdendritic regions. This qualitatively implies that the dendrite arms may be enriched in elements with more electrons (i.e., in other words, those heavy metals with high melting points in this case) than the interdendritic regions. Elemental segregation was further examined in NbTaTiVW using the EDS mapping on a small region (see Fig. 5 for details). For this example, W and Ta were mostly observed in the dendrite arm regions, while Nb, Ti, and V

enrichments were observed in the interdendritic regions. The EDS results qualitatively agree with the simulated composition profile generated for non-equilibrium solidification using CALPHAD.

The estimated bulk composition of the alloys (C_{aver}), the average composition of the dendritic arms (C_{dr}), and the average composition of the interdendritic regions (C_{idr}) are listed in Table 3. The measured C_{aver} deviates from the nominal concentration as a result of evaporation losses for each element during the melting process. In agreement with the qualitative analysis of the backscattered electron imaging (BEI) SEM images, the distribution of the elements in the as-cast alloys is inhomogeneous. Heavier metals with higher melting points are enriched in the dendrite arms, e.g., Ta and Nb for the NbTaTiV alloy; W and Ta for the NbTaVW and NbTaTiVW alloys. Conversely, lighter elements with lower melting points are enriched in the interdendritic regions.

3.4. Compressive properties and hardness

The true stress-strain (σ - ε) curves at room temperature (298 K) for the HEAs in this study are depicted in Fig. 6. The yield strength (i.e., $\sigma_{0.2}$ is defined as the stress at $\varepsilon=0.2\%$) for each HEA was determined and these values are listed in Table 2. The plastic fracture strains are 12% and 20% for the alloys NbTaVW and NbTaTiVW, respectively. Work hardening was observed in each alloy. Of particular note was HEA NbTaTiV, which did not fracture at the test compressive strain limit of 50%. An approximate work-hardening exponent, N , can be estimated using the Holloman equation:

$$\sigma = k\varepsilon^N, \quad (2)$$

where σ is the stress, k is the strengthening coefficient, and ε is the plastic strain [25]. The values of N were determined to be 0.12, 0.07, and 0.1 for alloys NbTaTiV, NbTaVW, and NbTaTiVW, respectively. Large N value often indicates more uniform plasticity. The values of k were determined to be 1468, 2160, and 1992 for the alloys NbTaTiV, NbTaVW, and NbTaTiVW, respectively. The average Vickers microhardness values (H_v) of the pure elements [13] and the alloys studied at ambient testing conditions are listed in Table 2.

4. Discussion

4.1. Phase formation

Several empirical parameters have been proposed to predict phase formation and structural stability in HEAs [26–29]. According to the classic Hume-Rothery rules, atomic size difference (δ), valence electron concentration (VEC), and electronegativity difference impact the formation of solid solutions. Besides these parameters, entropy and enthalpy of mixing (ΔH_{mix}) are the most critical phase formation factors for HEAs. The δ and ΔH_{mix} can be expressed as:

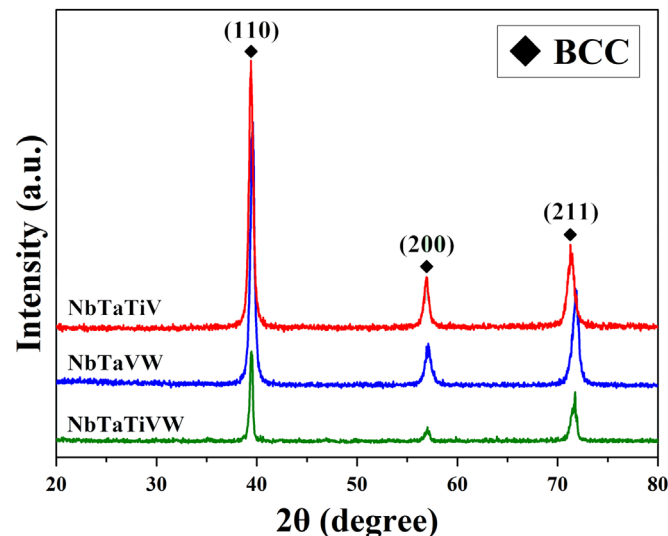


Fig. 3. X-ray diffraction patterns of the as-solidified HEAs NbTaTiV, NbTaVW, and NbTaTiVW. The corresponding index pattern for each is shown.

Table 2

The crystal lattice parameter (a), atomic radius (r), melting temperature (T_m), Vickers hardness (H_v), and yield strength ($\sigma_{0.2}$) in pure metals [13] are given. The calculated and experimental data for NbTaV-(Ti,W) HEAs are listed here.

Metal	Nb	Ta	Ti	V	W	NbTaTiV (Est/Exp)	NbTaVW (Est/Exp)	NbTaTiVW (Est/Exp)
a [Å]	3.301	3.303	3.276	3.039	3.158	3.2180/3.2193	3.2172/3.2186	3.2294/3.2244
r [Å]	1.46	1.46	1.47	1.34	1.39	–	–	–
T_m [K]	2741	3269	1930	2175	3680	2563/–	2979/–	2813/–
H_v [MPa]	1320	873	970	628	3430	961/3113 ± 46	1491/4830 ± 29	1409/4380 ± 34
$\sigma_{0.2}$ [MPa]	240	345	195	310	550	272/965	351/1530	327/1420

Note: the estimated values are calculated based on the bulk compositions from the EDX measurements.

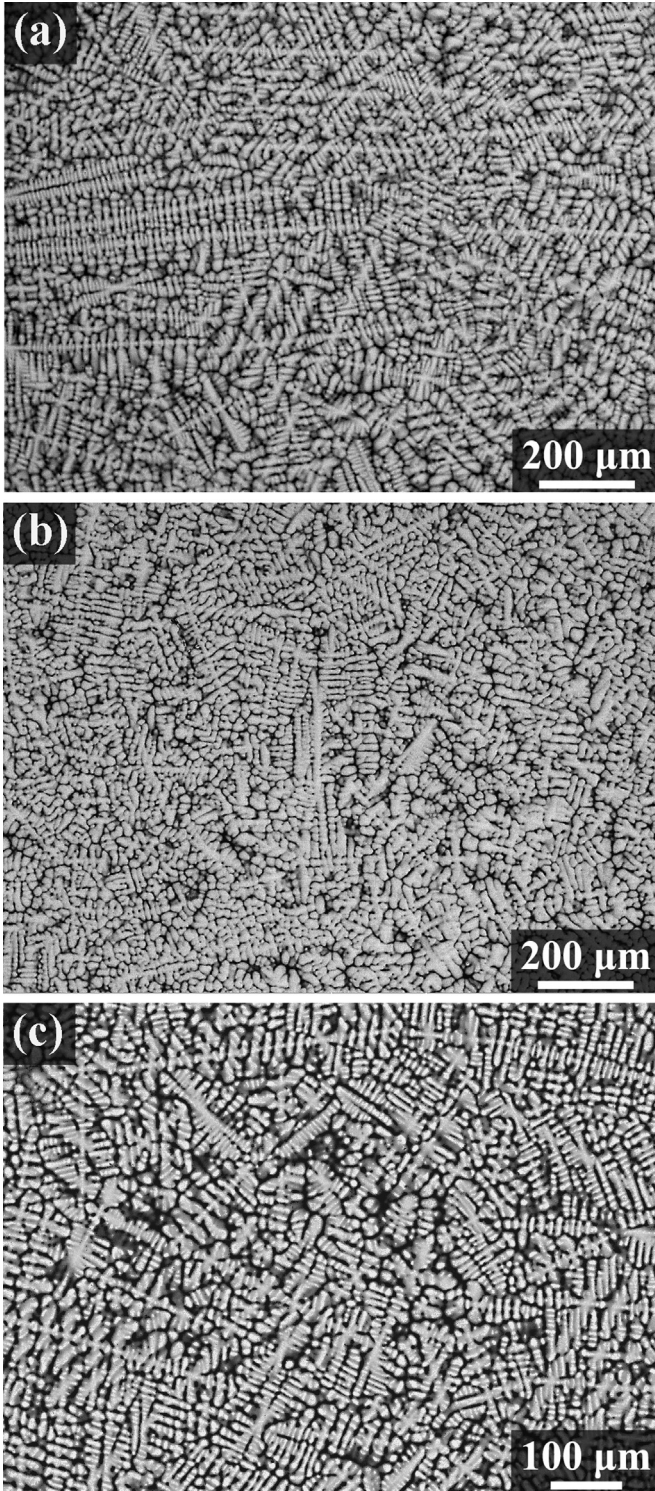


Fig. 4. Backscattered electron image SEM of polished cross-sections of HEAs: (a) NbTaTiV, (b) NbTaTiVW, and (c) NbTaTiVW.

$$\delta = 100\% \sqrt{\sum_{i=1}^n c_i (1 - r_i / \bar{r})^2}, \quad (3)$$

$$\Delta H_{\text{mix}} = \sum_{i=1, j \neq i}^n 4\omega_{ij} c_i c_j, \quad (4)$$

where $c_i(r_i)$ is the atomic fraction (the Goldschmidt atomic radius [13]) of the i^{th} component, and ω_{ij} is the enthalpy of mixing between i^{th} and j^{th} elements [30], which is based on the Miedema model [31]. The ΔH_{mix} reflects the tendency of forming stable intermetallic compounds while the δ characterizes the atomic size mismatch. Such an approach addresses the local elastic strain and the system topological instability [32]. Zhang et al. [26], for example, suggested that the criteria for forming random solid solutions alloys solidified from the melt were $-15 < \Delta H_{\text{mix}} < 5$ kJ/mol and $\delta < 5\%$.

Based on the concept of entropy-enthalpy competition, the Ω -parameter was proposed to be beneficial:

$$\Omega = \frac{T_m^{\text{mix}} \Delta S_{\text{mix}}}{|\Delta H_{\text{mix}}|}, \quad (5)$$

where $T_m^{\text{mix}} = \sum_{i=1}^n c_i(T_m)_i$ is the calculated average melting temperature. The ΔS_{mix} of HEAs is the ideal configurational entropy ($\Delta S_{\text{mix}}^{\text{conf}}$) in the solid/liquid state and is expressed as:

$$\Delta S_{\text{mix}}^{\text{conf}} = -R \sum_{i=1}^n c_i \ln c_i, \quad (6)$$

where R is the gas constant ($8.314 \text{ J K}^{-1} \text{ mol}^{-1}$). From available data, it was concluded that only solid solution phases were formed in many HEAs for $\Omega \geq 1.1$ and $\delta \leq 6.6\%$. For the current alloys, the relationship of Ω versus (δ versus ΔH_{mix}) is shown in Fig. 7. The data of other alloys are taken from Ref. [26].

In addition, Guo et al. [28] proposed that the FCC solid solution is stabilized for $VEC \geq 8$, and in the particular case of BCC solution, it is stabilized for $VEC < 6.87$. Coexistence of FCC and BCC phases is observed at VEC values between 6.87 and 8. The VEC for multi-component alloys can be defined as:

$$VEC = \sum_{i=1}^n c_i VEC_i, \quad (7)$$

where VEC_i is the VEC of the i^{th} element. The calculated VEC values for NbTaTiV, NbTaTiVW, and NbTaTiVW HEAs are 4.74, 5.21, and 5.02, respectively, each of which is much lower than the proposed threshold value of 6.87.

4.2. Thermodynamic properties analyses

The present research demonstrates that these empirical parameters are useful in predicting the phases that form during solidification. However, it should be noted that the Ω -parameter is calculated using the ideal configurational ΔS_{mix} and ΔH_{mix} for liquid phase as well as an estimated melting point. Consequently, the Ω -parameter may vary substantially if the thermodynamic properties of the solid phase and the liquidus temperature are used, as pointed out by Gao et al. [16]. Accordingly, the thermodynamic properties of the Nb-Ta-Ti-V-W system were further studied using CALPHAD.

The calculated enthalpies of mixing and entropies of mixing for both BCC and liquid phases (i.e., $\Delta H_{\text{mix}}^{\text{liq}}$, $\Delta H_{\text{mix}}^{\text{bcc}}$, $\Delta S_{\text{mix}}^{\text{liq}}$, and $\Delta S_{\text{mix}}^{\text{bcc}}$) are listed in the Table 1. NbTaTiV shows nearly zero $\Delta H_{\text{mix}}^{\text{liq}}$ and positive $\Delta H_{\text{mix}}^{\text{bcc}}$, and the entropies of mixing for both liquid and BCC phases equal the ideal configurational entropy. In contrast, NbTaTiVW and NbTaTiVW show apparent negative deviation from the ideal configurational entropy, which may be due to chemical ordering, chemical separation, and/or lattice phonon vibration. The enthalpies of mixing are negative for both liquid and BCC phases for NbTaTiVW and NbTaTiVW. Although the entropies of mixing are fairly close between liquid and BCC phases, there are noticeable differences in the enthalpies of mixing between liquid and BCC phases for each alloy. This is not surprising as a prior study by Gao et al. [16] also showed apparent difference between $\Delta H_{\text{mix}}^{\text{liq}}$ and

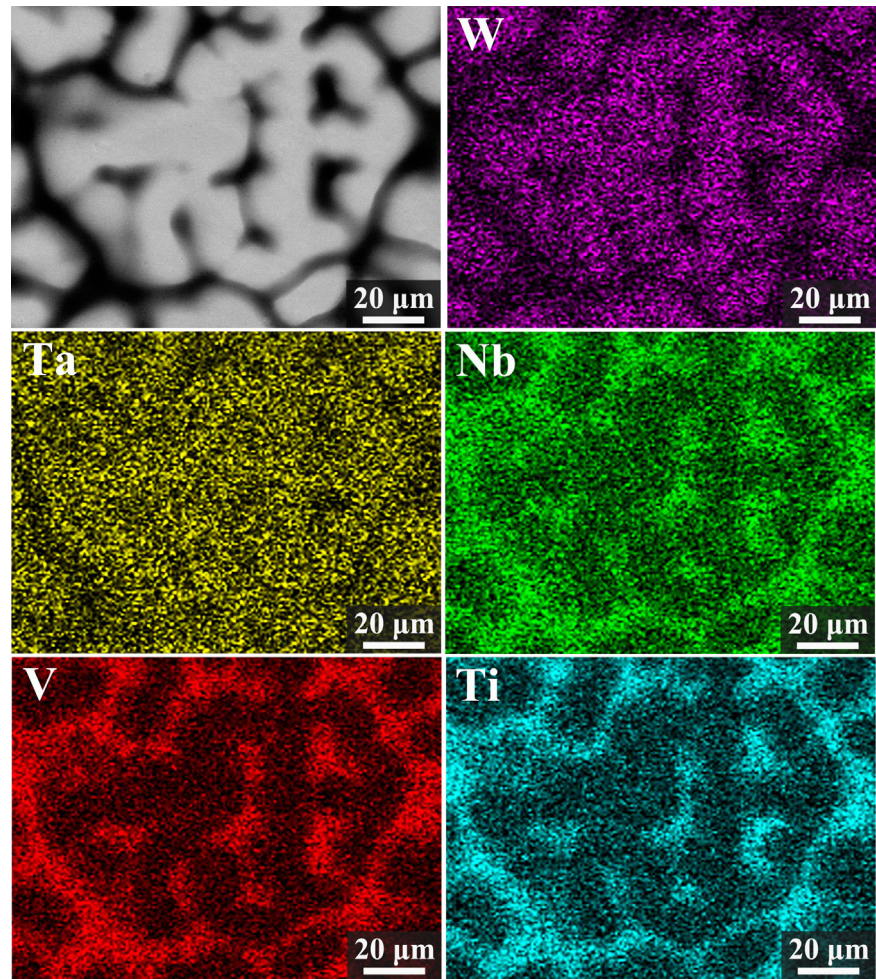


Fig. 5. Backscattered electron image SEM and EDS mapping of Nb, Ta, Ti, V, and W in the as-cast HEA NbTaTiVW.

Table 3
The estimated bulk composition (C_{aver}), the average composition of dendrite arms (C_{dr}), the average composition of interdendrite regions (C_{idr}), and the partition coefficient ($K=C_{dr}/C_{idr}$) for NbTaV-(Ti,W) HEAs.

Alloy	Concentrations (at.%)	Nb	Ta	Ti	V	W
NbTaTiV	C_{aver}	27.8	27.45	22.55	21.55	–
	C_{dr}	28.06	29.54	22.34	20.49	–
	C_{idr}	21.47	12.27	36.03	30.2	–
	$\Delta T_i=(T_m)_i - T_m^{mix}$	178	706	–633	–388	–
	$K=C_{dr}/C_{idr}$	1.31	2.41	0.62	0.68	–
NbTaVW	C_{aver}	28.50	29.65	–	20.67	21.18
	C_{dr}	26.82	32.05	–	15.11	26.02
	C_{idr}	33.95	25.19	–	29.12	11.74
	$\Delta T_i=(T_m)_i - T_m^{mix}$	–238	290	–	–804	701
	$K=C_{dr}/C_{idr}$	0.79	1.27	–	0.52	2.22
NbTaTiVW	C_{aver}	22.7	25.35	16.7	17.35	18
	C_{dr}	21.9	26.55	15.4	16.9	19.15
	C_{idr}	23.93	12.9	29.18	28.95	5.08
	$\Delta T_i=(T_m)_i - T_m^{mix}$	–72	456	–883	–638	867
	$K=C_{dr}/C_{idr}$	0.92	2.06	0.53	0.58	3.77

ΔH_{mix}^{bcc} for sixteen refractory BCC HEAs. In that study [16], both the sign and absolute values of ΔH_{mix}^{bcc} were not necessarily in accord with ΔH_{mix}^{liq} : For example, nine alloys exhibited opposite sign, while seven alloys show significant difference in their absolute values.

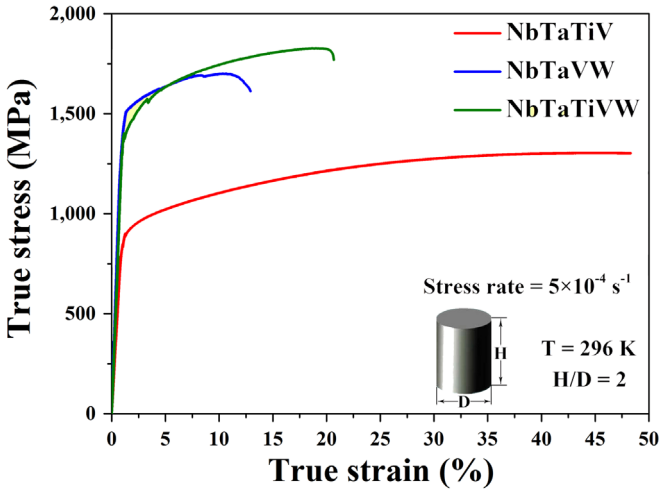


Fig. 6. Compressive true stress-strain curves for HEAs NbTaTiV, NbTaVW, and NbTaTiVW at room temperature.

To analyze the compositional dependence of the thermodynamic properties, the mixing behaviors in entropy, enthalpy, and Gibbs energy for the BCC phase for four pseudo-binaries of NbTaV-Ti, NbTaV-W, NbTaVW-Ti, and NbTaTiV-W at 1273 K were calculated. These results are shown in Fig. 8. The ΔS_{mix}^{bcc} for NbTaV-W, NbTaVW-Ti, and NbTaTiV-W pseudo-binaries lie well below their ideal mixing entropies, and thus they exhibit negative excess

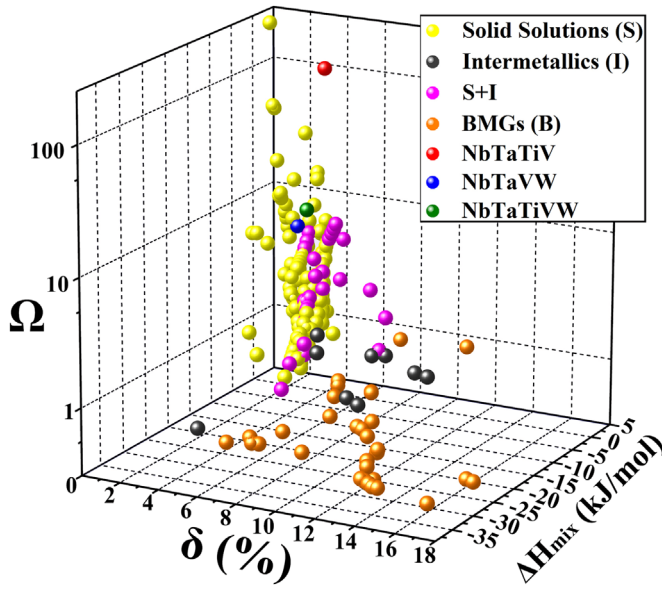


Fig. 7. A three-dimensional (3D) graph shows the relation of Ω versus (δ versus ΔH_{mix}) for NbTaTiV, NbTaVW, and NbTaTiVW HEAs.

entropy. In contrast, NbTaV-Ti behaves more like the ideal entropy of mixing. The $\Delta H_{\text{mix}}^{\text{bcc}}$ is positive for NbTaV-Ti system, and with the addition of Ti to NbTaVW, $\Delta H_{\text{mix}}^{\text{bcc}}$ increases. Conversely, with the addition of W to NbTaV or NbTaTiV, $\Delta H_{\text{mix}}^{\text{bcc}}$ decreases. The lowest $\Delta H_{\text{mix}}^{\text{bcc}}$ values are -7.8 kJ/mol for Nb_{16.2}Ta_{16.2}V_{16.2}W_{51.4} and -5.4 kJ/mol for Nb_{10.85}Ta_{10.85}Ti_{10.85}V_{10.85}W_{56.6}. The Gibbs energy of mixing combines the effect of $-\Delta S_{\text{mix}}^{\text{bcc}}$ and $\Delta H_{\text{mix}}^{\text{bcc}}$.

4.3. Microsegregation

The microsegregation of the elemental constituents demonstrates non-equilibrium solidification of the HEAs between the liquidus and solidus temperatures. Therefore, homogeneous distribution of the alloying elements in the growing solid phases is kinetically restricted [5]. The level of the element segregation is triggered by the different solidification temperatures associated with each element in the alloy system. The partition coefficient, $K = C_{\text{dr}}/C_{\text{idr}}$ [33], is commonly used to quantitatively evaluate the microsegregation of the alloying elements. The K value of an alloying element, which reflects the correlation of the excess/deficiency of the concentration in the dendrite arms, is often associated with its melting temperature (T_m). The elements with high

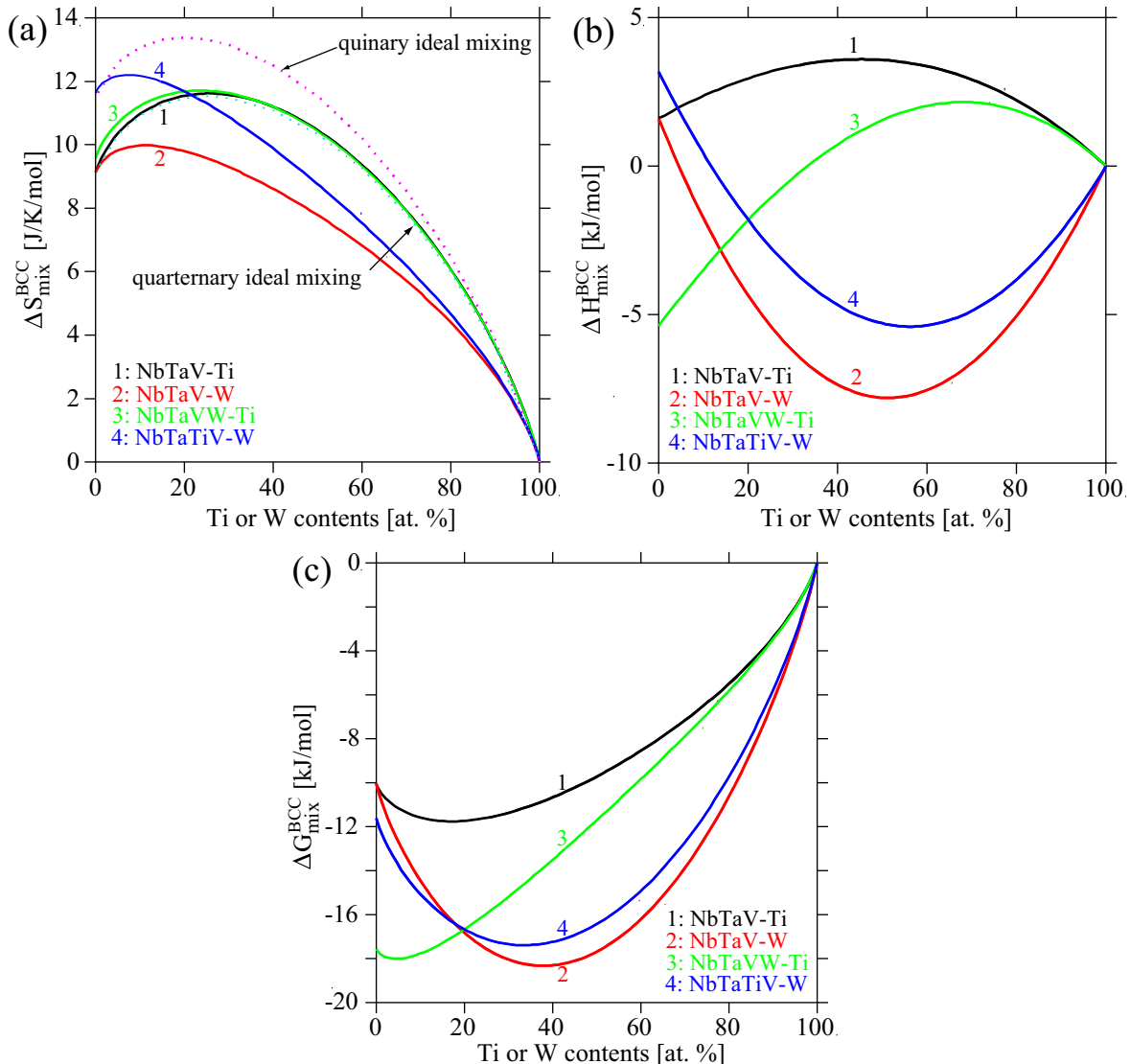


Fig. 8. Calculated (a) entropy of mixing, (b) enthalpy of mixing, and (c) Gibbs energy of mixing at $T = 1273$ K for four pseudo-binaries of NbTaV-Ti, NbTaV-W, NbTaVW-Ti, and NbTaTiV-W using TCNI7 database. The reference states used are the BCC phase at $T = 1273$ K.

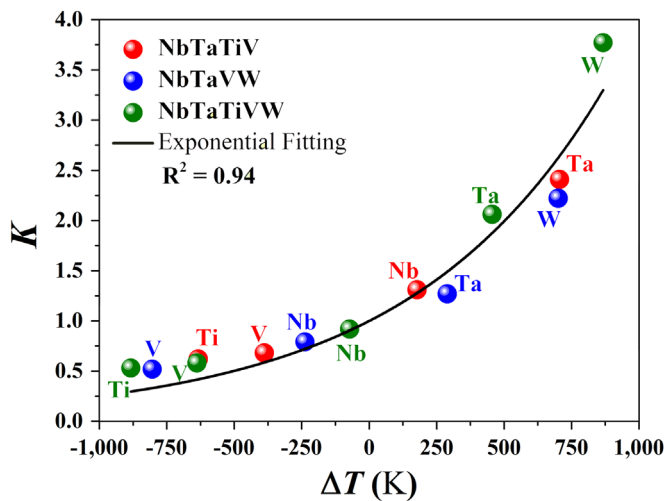


Fig. 9. The correlation between the partition coefficient, $K = C_{dr}/C_{idr}$, and the difference, $\Delta T = (T_m)_i - T_m^{mix}$.

melting temperature in the alloy will usually solidify first in the dendrite arms, constraining those elements with lower melting temperature to solidify later in the solidification process in the interdendritic regions. The correlation between K and temperature can be described as:

$$K = e^{0.0013\Delta T}, \quad (8)$$

where e is the natural base, and $\Delta T_i = (T_m)_i - T_m^{mix}$. Here, $(T_m)_i$ is the melting temperature of the i^{th} element, and T_m^{mix} is the calculated average melting temperature of the alloy.

The relationship between the K value of the i^{th} element (Ti, V, Nb, Ta, or W) in the present alloys and the temperature difference (see Fig. 9) shows that the segregation of the i^{th} element obeys a power-law distribution with ΔT_i , which was also observed in other refractory HEAs [34]. For the elements with $\Delta T_i > 0$, the partition coefficient is $K > 1$; for the elements with $\Delta T_i < 0$, the partition coefficient is $K < 1$. On the other hand, the predicted microsegregation of the NbTaTiVW alloy using the Scheil model via CALPHAD simulations are in qualitative agreement with experimental observation (see Fig. 2), and this indicates the usefulness of CALPHAD modeling for alloy design and solidification prediction. More realistic solidification of HEAs will involve both mobility and thermodynamic databases [20].

4.4. Room-temperature yield strength and hardness

In general, materials properties cannot be estimated using the ROM because they are sensitive to alloying elements and their concentrations. The ROM can be expressed as follow:

$$p = \sum_{i=1}^n c_i p_i, \quad (9)$$

where c_i and p_i are the atomic fraction and mechanical property of the i^{th} constituent element, respectively. The calculated average values of $\sigma_{0.2}^{mix}$ and H_v^{mix} are tabulated in Table 2, which are much smaller than the measured values. The results suggest that the mechanical properties of the three alloys disobey the simple ROM, and aspects of solid solution strengthening (SSS) may play a more dominant role than expected. Similar observations have been reported in other single-phase HEAs. The measured Vickers hardness for CoCrFeNi [35], MoNbTaV [36], MoNbTaW [5], MoNbTaVW [5], HfNbTaTiZr [37], MoNbTaTiVW [38], HfNbTaTiVZr [15], and CrMoNbTaVW [17] are 1.58, 4.5, 2.49, 3.37, 3.28, 3.40, 5.09, and

4.69 times the average values obtained from the ROM, respectively. However, little SSS is observed in the HCP GdHoLaTbY alloy [39] which behaves as an ideal solid solution because of the combination of extremely low $\Delta\chi$ (0.05) and zero ΔH_{mix} value, the lattice distortion is minimal, and thus, the solid solution strengthening is negligible.

5. Conclusions

In summary, three refractory HEAs, NbTaTiV, NbTaVW, and NbTaTiVW, were designed by CALPHAD modeling and synthesized by vacuum arc melting. A single BCC solid solution phase was observed in the as-cast alloys using a combination of XRD and SEM analyses. The following conclusions were reached:

- 1) While NbTaTiV is very ductile and does not fracture up to 50% strain, NbTaTiVW has a yield strength of 1420 MPa with a fracture plastic strain of 20%.
- 2) The Vickers microhardness values (and the yield strengths) of the HEAs are about 3 (and 3.5–4.4) times that those estimated from the rule of mixture, suggesting contribution from matrix solid solution strengthening.
- 3) The microsegregation behavior of the elements is closely related to their melting points, and the experimental results are consistent with CALPHAD prediction.
- 4) The entropies of mixing for both liquid and BCC phases for HEA NbTaTiV are predicted to be ideal. Conversely, HEAs NbTaVW and NbTaTiVW show negative departure from ideal.

Disclaimer

The computational modeling work presented in the paper is project was funded by the Department of Energy, National Energy Technology Laboratory, an agency of the United States Government, through a support contract with AECOM. Neither the United States Government nor any agency thereof, nor any of their employees, nor AECOM, nor any of their employees, makes any warranty, expressed or implied, or assumes any legal liability or responsibility for the accuracy, completeness, or usefulness of any information, apparatus, product, or process disclosed, or represents that its use would not infringe privately owned rights. Reference herein to any specific commercial product, process, or service by trade name, trademark, manufacturer, or otherwise, does not necessarily constitute or imply its endorsement, recommendation, or favoring by the United States Government or any agency thereof. The views and opinions of authors expressed herein do not necessarily state or reflect those of the United States Government or any agency thereof.

Acknowledgments

The authors would like to acknowledge the financial support of National Natural Science Foundation of China (Nos. 51371122 and 51501123), the Program for the Innovative Talents of Higher Learning Institutions of Shanxi (2013), the Youth Natural Science Foundation of Shanxi Province, China (Nos. 2015021005 and 2015021006), and the financial support from State Key Lab of Advanced Metals and Materials (No. 2015-Z07). The modeling work was funded by the Cross-Cutting Technologies Program at the National Energy Technology Laboratory (NETL)—Strategic Center for Coal, managed by Robert Romanosky (Technology Manager) and Charles Miller (Technology Monitor). The Research

was executed through NETL's Office of Research and Development's Innovative Process Technologies (IPT) Field Work Proposal. Research performed by AECOM Staff was conducted under the RES contract DE-FE-0004000.

References

- [1] R.O. Ritchie, *Nat. Mater.* 10 (2011) 817–822.
- [2] T. Chookajorn, H.A. Murdoch, C.A. Schuh, *Science* 337 (2012) 951–954.
- [3] J.W. Yeh, S.K. Chen, S.J. Lin, J.Y. Gan, T.S. Chin, T.T. Shun, et al., *Adv. Eng. Mater.* 6 (2004) 299–303.
- [4] B. Cantor, I.T.H. Chang, P. Knight, A.J.B. Vincent, *Mater. Sci. Eng. A* 375 (2004) 213–218.
- [5] O.N. Senkov, G.B. Wilks, D.B. Miracle, C.P. Chuang, P.K. Liaw, *Intermetallics* 18 (2010) 1758–1765.
- [6] A. Takeuchi, K. Amiya, T. Wada, K. Yubuta, W. Zhang, *JOM* 66 (2014) 1984–1992.
- [7] M.C. Gao, B. Zhang, S.M. Guo, J.W. Qiao, J.A. Hawk, *Metall. Mater. Trans. A* 47A (2016) 3322–3332.
- [8] Y. Zhang, T.T. Zuo, Z. Tang, M.C. Gao, K.A. Dahmen, P.K. Liaw, et al., *Prog. Mater. Sci.* 61 (2014) 1–93.
- [9] M.C. Gao, J.W. Yeh, P.K. Liaw, Y. Zhang, *High-Entropy Alloys: Fundamentals and Applications*, 1st ed., Springer International Publishing, Cham, Switzerland, 2016.
- [10] M.C. Gao, Ö.N. Doğan, P. King, A.D. Rollett, M. Widom, *JOM* 60 (2008) 61–65.
- [11] O.N. Senkov, G.B. Wilks, J.M. Scott, D.B. Miracle, *Intermetallics* 19 (2011) 698–706.
- [12] O.N. Senkov, S.V. Senkova, D.M. Dimiduk, C. Woodward, D.B. Miracle, *J. Mater. Sci.* 47 (2012) 6534.
- [13] (<http://en.wikipedia.org/wiki/Element>).
- [14] B. Sundman, B. Jansson, J.-O. Andersson, *Calphad* 9 (1985) 153–190.
- [15] M.C. Gao, B. Zhang, S. Yang, S.M. Guo, *Metall. Mater. Trans. A* 47A (2016) 3333–3345.
- [16] M.C. Gao, C.S. Carney, Ö.N. Doğan, P.D. Jablonksi, J.A. Hawk, D.E. Alman, *JOM* 67 (2015) 2653–2669.
- [17] B. Zhang, M. Gao, Y. Zhang, S.M. Guo, *Calphad* 51 (2015) 193–201.
- [18] C. Zhang, F. Zhang, S. Chen, W. Cao, *JOM* 64 (2012) 839–845.
- [19] M.C. Gao, D.E. Alman, *Entropy* 15 (2013) 4504–4519.
- [20] C. Zhang, M.C. Gao, Chapter 12: CALPHAD Modeling of High-Entropy Alloys, in *High-Entropy Alloys: Fundamentals and Applications*, 1st ed., M.C. Gao, J.W. Yeh, P.K. Liaw, Y. Zhang (Eds.), Springer International Publishing, Cham, Switzerland, 2016.
- [21] F. Zhang, C. Zhang, S. Chen, J. Zhu, W. Cao, U. Kattner, *Calphad* 45 (2014) 1–10.
- [22] G.H. Gulliver, *J. Inst. Met.* 9 (1913) 120–157.
- [23] E. Scheil, *Z. Met.* 34 (1942) 70–72.
- [24] L. Vegard, *Z. Phys.* 5 (1921) 17–26.
- [25] J.H. Hollomon, *AIIME Trans.* 12 (1945) 1–22.
- [26] X. Yang, Y. Zhang, *Mater. Chem. Phys.* 132 (2012) 233–238.
- [27] S. Guo, C.T. Liu, *Prog. Nat. Sci: Mater. Int.* 21 (2011) 433–446.
- [28] S. Guo, C. Ng, J. Lu, C.T. Liu, *J. Appl. Phys.* 109 (2011) 103505.
- [29] Y. Zhang, Y.J. Zhou, J.P. Lin, G.L. Chen, P.K. Liaw, *Adv. Eng. Mater.* 10 (2008) 534–538.
- [30] A. Takeuchi, A. Inoue, *Mater. Trans.* 46 (2005) 2817–2829.
- [31] G. Gulliver, *J. Inst. Metals* 9 (1913) 120–157.
- [32] T. Egami, Y. Waseda, *J. Non Cryst. Solids* 64 (1984) 113–134.
- [33] M.C. Flemings, New York, 1974.
- [34] A. Clauset, C.R. Shalizi, M.E. Newman, *SIAM Rev.* 51 (2009) 661–703.
- [35] T.T. Shun, L.Y. Chang, M.H. Shiu, *Mater. Charact.* 70 (2012) 63–67.
- [36] H.W. Yao, J.W. Qiao, M.C. Gao, J.A. Hawk, S.G. Ma, H.F. Zhou, *Entropy* 18 (2016) 189.
- [37] O.N. Senkov, J.M. Scott, S.V. Senkova, D.B. Miracle, C.F. Woodward, *J. Alloy. Compd.* 509 (2011) 6043–6048.
- [38] B. Zhang, M.C. Gao, Y. Zhang, S. Yang, S.M. Guo, *Mater. Sci. Technol.* 31 (2015) 1207–1213.
- [39] Y.J. Zhao, J.W. Qiao, S.G. Ma, M.C. Gao, H.J. Yang, M.W. Chen, et al., *Mater. Des.* 96 (2016) 10–15.